

Workflow documentation

ResearchOS — AI-assisted research workflow

CORE PIPELINE



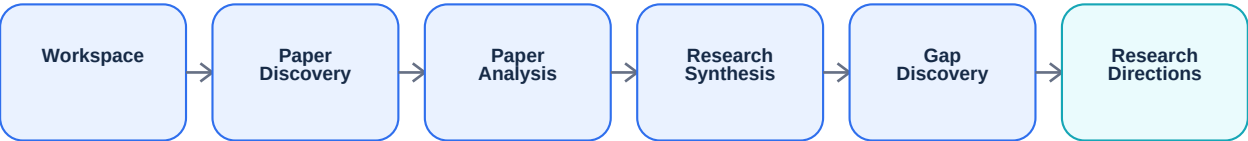
Prepared from the implemented n8n workflow definitions
and project workflow notes supplied for ResearchOS.

Submission-ready technical documentation

1. Executive Summary

ResearchOS is a modular, event-driven research automation platform that uses n8n to orchestrate research workflows, Supabase as a centralized data layer, academic APIs for paper discovery, and OpenAI GPT-5-mini for structured research reasoning. The workflow architecture progressively transforms a research question into a structured body of evidence, a cross-paper synthesis, identified research gaps, and concrete research directions.

The implemented workflow set is deliberately separated into independently callable stages. Each stage validates its inputs, performs a focused operation, validates the resulting output where appropriate, and returns a structured JSON response. This design improves maintainability, testability, and integration with a future frontend.



Design principle

ResearchOS treats AI output as structured, constrained reasoning rather than unrestricted text generation. The workflows instruct the model not to invent papers, citations, evidence, results, or unsupported factual claims.

2. System Architecture

ResearchOS follows a modular architecture in which specialized workflows communicate through structured data and a centralized research database. The automation layer is implemented in n8n, while external services provide academic retrieval, AI processing, and future communication/storage capabilities.

Layer	Responsibility
User / Interface	Collects research topics, questions, keywords, interests and search preferences.
Workflow Automation — n8n	Orchestrates independent workflows, validation branches, API calls, AI calls and database operations.
AI Intelligence	Performs paper analysis, synthesis, gap identification and research-direction generation using constrained prompts.
Research Data — Supabase	Stores workspaces, papers and structured paper analyses as the persistent source of truth.
External Services	Includes arXiv, Semantic Scholar and OpenAI; project notes also identify Gmail and Google Drive as integrated/future service layers.

Primary data entities

Entity	Role	Key information
workspaces	Research context	user_id, workspace_name, research_topic, research_question, keywords, search_preferences
papers	Discovered literature	title, authors, abstract, dates, identifiers, URLs, source, relevance, analysis status
paper_analysis	Structured AI knowledge	summary, contributions, methodology, datasets, models, limitations, future work, keywords

3. End-to-End Workflow Interaction

The project workflow notes describe an event-driven lifecycle in which a research topic becomes the central context for subsequent research operations. The implemented n8n workflows expose the individual stages through webhook endpoints, allowing the stages to be called independently while preserving a consistent data flow.



Important distinction
The project notes describe the broader product vision, including a weekly research monitor and Gmail notifications. This document focuses on the substantive workflow JSON implementations supplied for workspace creation, paper discovery, paper analysis, research synthesis, research-gap discovery and research-direction generation.

4. Workflow 1 — Workspace Creation

Purpose

Creates the research workspace that acts as the central context for a user's research topic. The workflow receives the user's initial research information, validates required fields, stores the workspace in Supabase, and returns the generated workspace ID.

Webhook endpoint

POST /webhook/researchos/workspace-create

Workflow sequence



Input fields

Field	Type	Required	Purpose
user_id	number	Yes	Associates the workspace with a user.
workspace_name	string	Yes	Human-readable workspace name.
research_topic	string	Yes	Main topic being investigated.
research_question	string	Yes	Question guiding the research.
keywords	array	No	Search terms for literature discovery.
search_preferences	object	No	Optional search configuration.

Implementation details

- The webhook extracts the six workspace fields from the request body.
- Required validation checks user ID, workspace name, research topic and research question.
- Valid input is inserted into the Supabase **workspaces** table.
- A successful creation returns HTTP **201** and the newly created workspace ID.
- Invalid input returns HTTP **400** with a validation message.

5. Workflow 2 — Paper Discovery

Purpose

Discovers relevant academic literature using both arXiv and Semantic Scholar. The workflow normalizes results into a common paper representation, merges the sources, removes duplicates, and stores the resulting records in Supabase.

Webhook endpoint

POST /webhook/researchchos/papers/discover

Workflow sequence



Normalization and deduplication

The workflow creates a common structure for papers from different sources. Records can include title, authors, abstract, publication date, venue, arXiv ID, Semantic Scholar ID, DOI, PDF URL, paper URL, source, citation count, relevance score, analysis status and indexing status.

- Search results from arXiv and Semantic Scholar are merged before persistence.
- Duplicate detection uses stable identifiers such as DOI, arXiv ID and Semantic Scholar ID, with title/year information used as a fallback.
- A relevance score is calculated from keyword matches against the title and abstract.
- Stored papers begin with an analysis status of **pending** and an indexed flag of **false**.

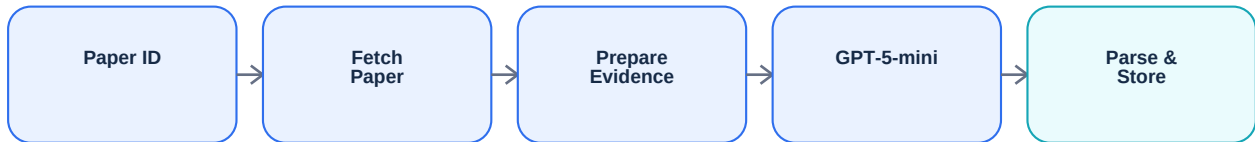
6. Workflow 3 — Paper Analysis

Purpose

Analyzes a stored paper using GPT-5-mini and converts its available metadata and abstract into structured research knowledge. The resulting analysis is persisted in the **paper_analysis** table for later synthesis.

Webhook endpoint

POST /webhook/researchos/papers/analyze



AI output schema

Field	Meaning
summary	Concise description of the paper based only on supplied information.
key_contributions	Contributions supported by the paper metadata/abstract.
methodology	Method or approach identifiable from the supplied text.
datasets	Datasets explicitly identified in the supplied information.
models_used	Models explicitly identified in the supplied information.
limitations	Supported limitations; otherwise left empty.
future_work	Future-work items supported by the supplied information.
keywords	Relevant terms extracted from the paper.
research_gaps	Potential gaps indicated by the supplied information.

Reliability controls

- The prompt instructs the model to use only the supplied metadata and abstract.
- The model is told not to invent experiments, results, datasets, models, citations, authors or limitations.
- Missing information is represented with empty arrays or strings rather than fabricated content.
- The AI output is parsed as JSON before being transformed into a database record.
- The workflow returns explicit errors for invalid input, missing papers, invalid AI output and database insertion failures.

7. Workflow 4 — Research Synthesis

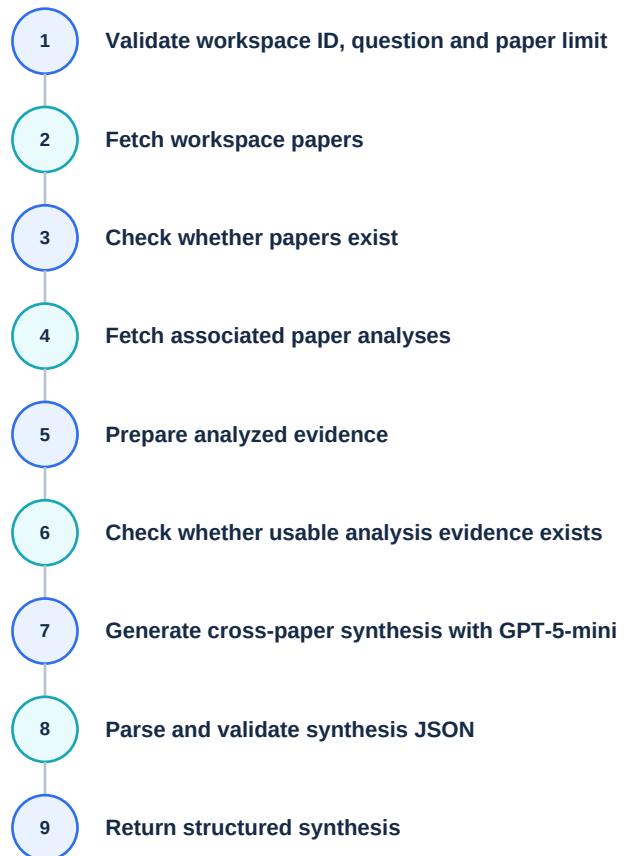
Purpose

Combines analyses from multiple papers within a workspace and generates a cross-paper synthesis addressing the research question. The workflow intentionally asks the model to synthesize evidence rather than simply summarize papers one by one.

Webhook endpoint

POST /webhook/researchos/research/synthesize

Workflow sequence



Synthesis schema

Field	Purpose
research_question	Research question addressed by the synthesis.
overview	High-level synthesis of the evidence.
key_themes	Cross-paper themes with supporting paper IDs.
major_findings	Major findings supported by the analyzed papers.
comparison	Comparison of evidence or approaches across papers.
research_gaps	Gaps visible in the available evidence.
limitations	Limitations of the evidence or synthesis.
conclusion	Overall conclusion grounded in the evidence.
papers_considered	Actual paper IDs used for synthesis.

Evidence-grounding rule

The synthesis prompt explicitly requires the model to distinguish directly supported findings from reasonable cross-paper synthesis and from evidence gaps. If the evidence is insufficient, the model is instructed to say so rather than fill the gap with invention.

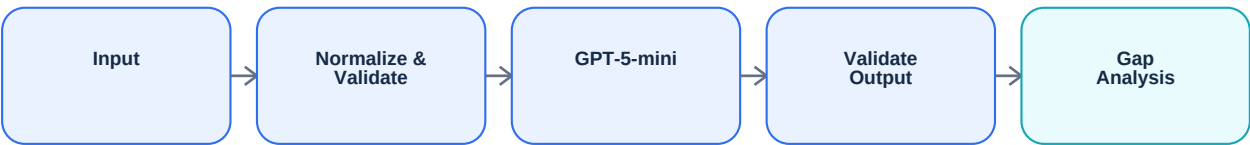
8. Workflow 5 — Research Gap Discovery

Purpose

Identifies research gaps from an already-generated research synthesis. It is intentionally downstream of synthesis so that gap identification is anchored to the evidence available to ResearchOS rather than to an unrestricted new literature review.

Webhook endpoint

POST /webhook/researchos/research/gaps



Required input

- **workspace_id** — positive numeric workspace identifier.
- **research_question** — non-empty research question.
- **synthesis** — existing research synthesis object.

Output structure

Field	Description
gap_summary	Concise overview of the major research gaps.
research_gaps	Objects containing the gap, evidence supporting it, and why it matters.
research_opportunities	Potential research directions suggested by the identified gaps.

The workflow validates that the AI result contains a non-empty **research_gaps** array before returning success. Invalid input produces HTTP 400; failed AI analysis produces HTTP 500.

9. Workflow 6 — Research Direction Generation

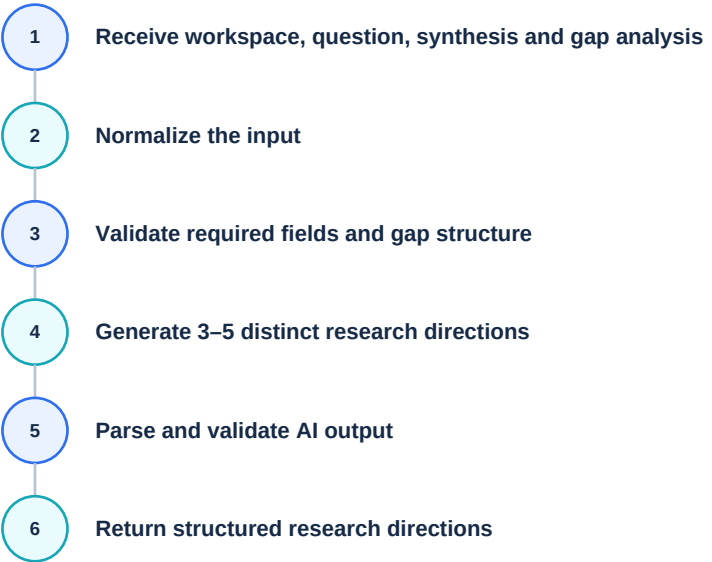
Purpose

Transforms identified research gaps into concrete, researchable directions. A direction is treated as a proposed way to investigate a gap, not as an established factual conclusion.

Webhook endpoint

POST /webhook/researchos/research/directions

Workflow sequence



Direction schema

Field	Purpose
direction_summary	Concise description of the overall direction landscape.
title	Short research direction title.
research_question	Specific question that could be investigated.
rationale	Why the direction follows from the identified gaps.
expected_contribution	Potential contribution of the proposed research.

Generation constraint

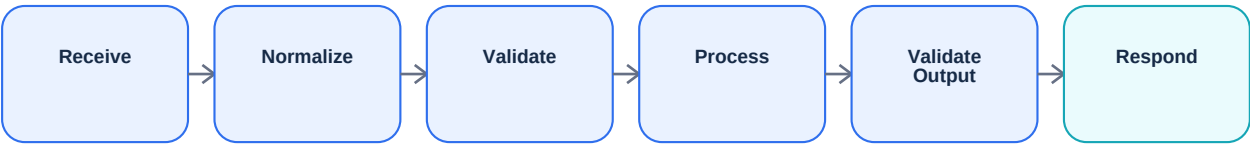
The model is instructed to reason only from the supplied research question, synthesis and gap analysis. It must not perform a new literature review or invent papers, citations, evidence or experimental results.

10. Validation, Error Handling & API Responses

Validation is a recurring design pattern across the workflows. Inputs are normalized before use, required fields are checked, and AI outputs are parsed and validated before being persisted or returned.

Condition	Typical response	Example workflow behaviour
Valid request	200	Returns structured success payload.
Workspace creation succeeds	201	Returns workspace ID and creation message.
Invalid required input	400	Stops processing and returns validation error.
Requested paper/workspace data not found	404	Returns resource-not-found response.
AI parsing or database failure	500	Returns processing/storage error.

Validation pattern



11. Reproducibility & Deployment Notes

The workflows are distributed as n8n JSON exports. Each export preserves the node configuration, connections, expressions, prompts and processing logic needed to reconstruct the workflow in an n8n environment.

- Import the supplied workflow JSON files into n8n.
- Configure the required Supabase credential and ensure the relevant database tables exist.
- Configure the OpenAI credential used by the AI nodes.
- Configure external academic API access where required by the Paper Discovery workflow.
- Activate the workflows and call their webhook endpoints with the documented JSON payloads.
- Execute the stages in sequence when using the complete research lifecycle.

Workflow endpoints at a glance

Workflow	Endpoint
Workspace Creation	/webhook/researchchos/workspace-create
Paper Discovery	/webhook/researchchos/papers/discover
Paper Analysis	/webhook/researchchos/papers/analyze
Research Synthesis	/webhook/researchchos/research/synthesize
Research Gap Discovery	/webhook/researchchos/research/gaps
Research Direction Generation	/webhook/researchchos/research/directions

12. Implementation Status & Scope

The supplied workflow set contains six substantive workflow implementations covering the core research lifecycle. The final research-direction workflow is represented by a complete exported definition containing its webhook, validation, GPT-5-mini generation, output parsing, validation and response branches.

Workflow	Implementation status
Workspace Creation	Implemented in exported n8n workflow.
Paper Discovery	Implemented in exported n8n workflow.
Paper Analysis	Implemented in exported n8n workflow.
Research Synthesis	Implemented in exported n8n workflow.

Workflow	Implementation status
Research Gap Discovery	Implemented in exported n8n workflow.
Research Direction Generation	Implemented in exported n8n workflow; final live execution was not fully verified after the n8n Cl

Documentation scope

This document distinguishes the implemented workflow definitions from broader product-vision notes. Features such as continuous weekly monitoring, Gmail notification delivery and future recommendation/paper-comparison capabilities are described in the project notes as broader or future functionality and are not presented here as fully implemented core workflows.

13. Conclusion

ResearchOS implements a clear research-automation pipeline: **Create** → **Discover** → **Analyze** → **Synthesize** → **Identify Gaps** → **Generate Directions**. The modular design separates retrieval, storage, AI reasoning and validation responsibilities, making the system easier to test and extend.

The most important architectural characteristic is the progressive refinement of evidence. Papers are first retrieved and normalized, then converted into structured analyses, synthesized across multiple sources, examined for gaps, and finally transformed into proposed research directions. This provides a coherent bridge from literature discovery to research ideation while retaining explicit safeguards against unsupported AI-generated claims.

Final workflow model

Research question → academic evidence → structured paper intelligence → cross-paper synthesis → evidence-grounded gaps → concrete research directions.